6 **(a)** **(i)** $^{220}_{86}Rn \rightarrow ^{216}_{84}Po + ^{4}_{2}He + \gamma$

total proton and nucleon numbers equal on both sides B1

equation include photon B1

(ii) 1.0 MeV = $1 \times 10^{6} \times 1.6 \times 10^{-19} = 1.6 \times 10^{-13}$ J A1

(iii) **1.** energy released
= 6.29 MeV + 0.55 MeV (= 6.84 MeV = 1.094×10^{-12} J) C1

$E = mc^2$
$m = 1.094 \times 10^{-12} / (3 \times 10^{8})^{2} = 1.2 \times 10^{-29}$ kg A1

2. $E = hf = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$
wavelength
= $(6.63 \times 10^{-34} \times 3 \times 10^{8}) / (0.55 \times 1.6 \times 10^{-13})$ M1

= 2.3×10^{-12} m A1

(b) **(i)** N_0 is initial number of C-14 atoms, N_t is number of C-14 atoms now and N_{12} be number of C-12 atoms. So

$N_t = N_0 \exp\left(-\frac{\ln 2}{t_{1/2}} t\right)$ C1

$\frac{N_t}{N_{12}} = \frac{N_0}{N_{12}} \exp\left(-\frac{\ln 2}{t_{1/2}} t\right)$
$\frac{1}{8.6 \times 10^{10}} = \frac{1}{3.3 \times 10^{10}} \exp\left(-\frac{\ln 2}{5700} t\right)$ C1

t = 7900 years A1

(ii) **1.** mass of a Carbon-14 ion different from Carbon-12 ion B1

same magnetic force on each atom because they have same speed and charge B1

so, radius of curvature of path of each type of ion in magnetic field is different

2. method enables determination of amount of Carbon-14 exactly B1

instead of measuring the radioactivity which is probabilistic in nature B1

7 **(a)** A photon is a quantum (or packet/ discrete bundle) of electromagnetic radiation (energy) B1